# Exit Ticket · reflect and retrieve

## Exit Ticket

A loop using i<=Length can safely read a[i] for every i.

## Use key vocabulary from memory

1. Explain logic error in your own words.

2. Explain boundary in your own words.

3. Explain trace in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Lesson 086 · Debugging Unfamiliar Algorithms

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What is a loop invariant?

### Q2

What is a boundary test?

### Q3

What return value commonly indicates a failed indexed search?

### Q4

In array [2,4,6], set a[1]=a[0]+a[2]. Give the final array.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Observe — Trace before changing code: list variables, loop indexes, condition and expected return.

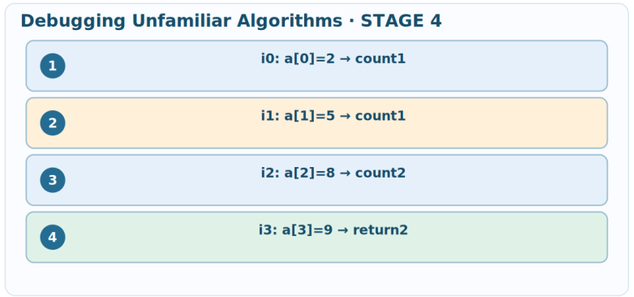
## Core idea 2

Explain — An off-by-one fault often comes from <= Length instead of < Length; an empty array must also be considered.

## Core idea 3

Evaluate — Correct code by explaining the cause and showing a fresh test, not just copying a working version.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Develop four distinct tests for a function counting even values, with expected answers and reasons. [8]

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Trace CountEven([2,5,8,9]) with count after each item. [6]

## Question 2

Explain why i<=a.Length is unsafe in a loop indexing a[i], and repair. [6]

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

logic error — Program runs but produces wrong answer.

boundary — Edge of valid range.

trace — Record changing states.

debugging — Finding and correcting faults.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

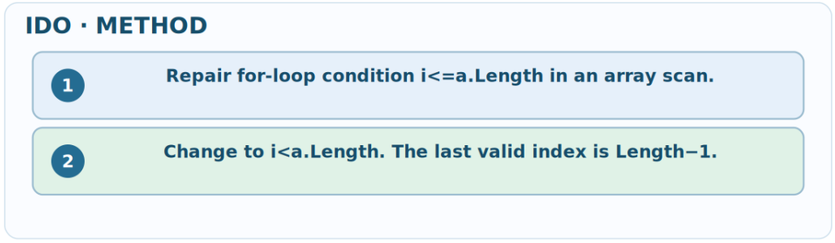
## Problem

Repair for-loop condition i<=a.Length in an array scan.

## Follow the worked method

1. Model response — For length4, valid indexes0–3. Checking a[4] would throw.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Model response
Initialising to1 incorrectly counts an item before any check.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.